Pune Vidyarthi Griha’s College of Engineering,
Technology and Management

Project Review 2

OIL SLICK DETECTION and DRIFT PREDICTION USING SATELLITE IMAGERY

Guide:
Prof. M.R. Apsangi

By:
Paras Ningune - 3078
Sakshi Patil - 3020
Soham Mane - 3021
Siddhant Pote - 3037

Oil Slicks, Their Impacts & Importance of Prediction

- **What are Oil Slicks?**
  - Thin layers of oil floating on the ocean surface, often from spills, leaks, or discharges.
- **Impacts**
  - Harm marine life and fragile ecosystems (corals, mangroves, seabirds).
  - Cause economic losses in fisheries, aquaculture, and coastal tourism.
- **Why Detection & Prediction Matters**
  - Enables early response to minimize ecological and economic damage.
  - Forecasting drift paths supports containment, cleanup, and risk assessment.

Purpose

To design an efficient pipeline for detecting oil slicks and predicting their drift using satellite imagery.

Objective

- Apply image processing & deep learning to reduce false alarms
- Improving the prediction accuracy and generalizing the model
- Integrate environmental data (currents, wind, tides, waves, temperature) for drift prediction
- Forecast slick movement and impacted areas through predictive modeling
- Develop a unified, scalable system for real-time monitoring and response

Scope

- Focus on satellite-based monitoring, mainly Synthetic Aperture Radar (SAR).
- Compare classical thresholding, ML based and DL based approaches for oil slick detection.
- Integration of drift prediction models with detection to simulate slick movement.
- Exploring datasets and Environmental data.

Hardware

- Standard workstation with GPU support. (Minimum RTX4050)

Data

- Dataset: Sentinel-1 SAR Oil spill image dataset.
- Weather Data: OpenWeatherMap API

Software / Libraries

- Operating System: Ubuntu
- Python (TensorFlow, PyTorch, Scikit-learn, OpenCV).
- QGIS and GNOME.

Satellite

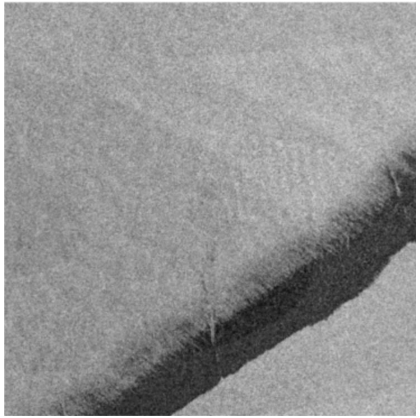
- **Synthetic Aperture Radar (SAR)** is a radar sensor used in remote sensing.
- It sends microwave signals towards earth and measures the echoes that bounce back.
- Due to its movement it creates a Synthetic Aperture that gives high resolution images.

Satellite	Sensor Type	Spatial Resolution	Revisit Time
Sentinel-1	C-Band SAR	~ 10m	6-12 Days

Polarization	Meaning	Modes
VV	Transmit Vertical, Receive Vertical	Single Polarization
HH	Transmit Horizontal, Receive Horizontal	Single Polarization
HV	Transmit Horizontal, Receive Vertical	Dual Polarization (HH + HV)
VH	Transmit Vertical, Receive Horizontal	Dual Polarization (VV + VH)

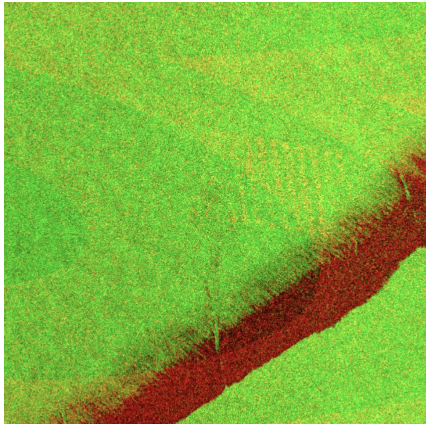
SAR Imagery

- If only one polarization is considered -> grayscale images.



HH

- If we combine multiple polarizations -> multi-band images



VV
+
VH

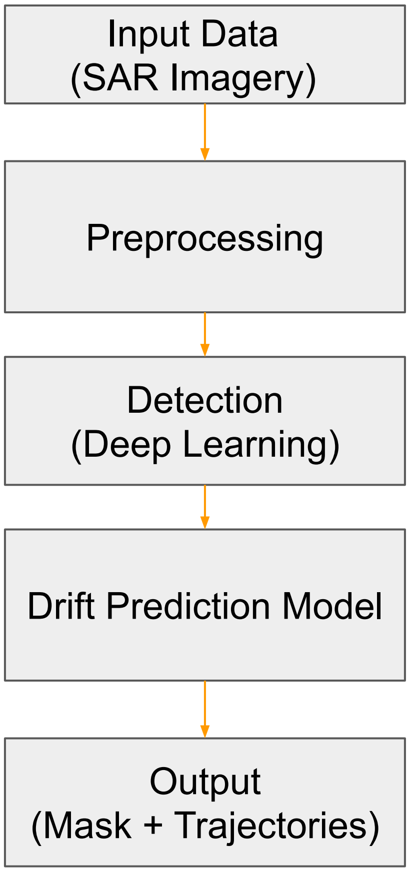
System Overview

Proposed System:

1. Oil detection using SAR imagery and DL.
2. Integrating environmental data for drift prediction (wind, wave current).
3. Drift modeling using hybrid (Lagrangian + Eulerian) and probabilistic models.

Expected Outcomes:

1. High-accuracy oil slick detection with reduced false positives.
2. Reliable drift forecasts.
3. Generalized model which can be used in any region of world.



Flow Chart

DL Models

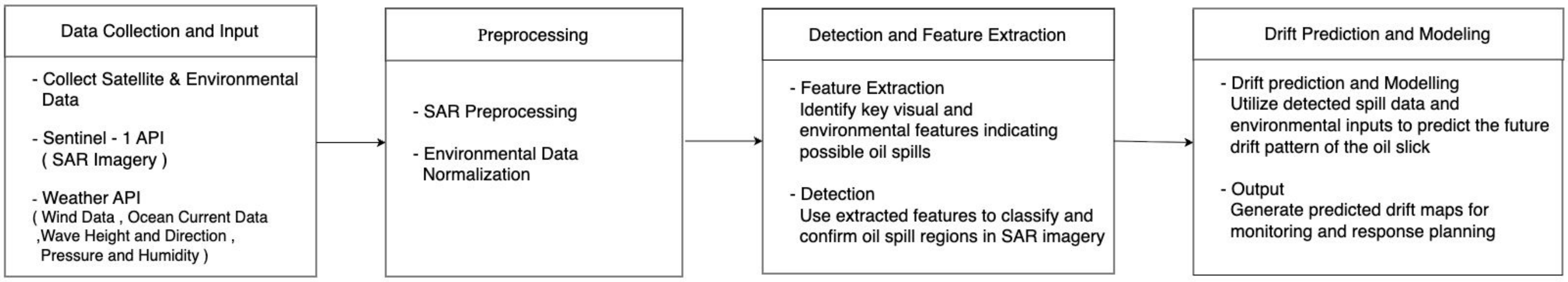
Aspects	CNN	U-Net	SegNet
Architecture	Learn patterns and features using convolution layers.	Encoder - Decoder CNN with skip connections.	Encoder - Decoder CNN with pooling indices for boundary reconstruction.
Strengths	- Captures local features effectively. - Highly scalable and can be trained on large datasets	- High segmentation accuracy. - Can be used across different sea states.	- Efficient in memory usage - Highly scalable - Light compared to U-nets
Weaknesses	- Requires large size datasets - They are more prone to overfitting especially in case of small datasets	- Requires large labelled datasets. - Requires use of heavy GPU's.	- Slightly less accurate than U-Nets. - Struggles with small size slicks.
Suitability	Early stage detection, patch detections.	Pixel-wise segmentation of slick boundaries.	Real time segmentation when resources are limited.
Citation	[6]	[8]	[7]

Literature Review

Paper Reference	Approach	Key Ideas	Pros	Cons
[4] Detection of oil slicks in SAR satellite images using Otsu-Bradley's thresholding method. [10] Textural Features for Image Classification.	Classical	Threshold & GCLM	Fast, Cheap	Noise Sensitive, False Alarms
[6] Oil Spill Detection using Convolutional Neural Networks and Sentinel-1 SAR Imagery. [7] Dark Spot Detection in SAR Images of Oil Spill Using Segnet. [8] U-Net: Convolutional Networks for Biomedical Image Segmentation.	DL	CNN, UNet Segmentation	High accuracy, Auto learns	Expensive, data-hungry

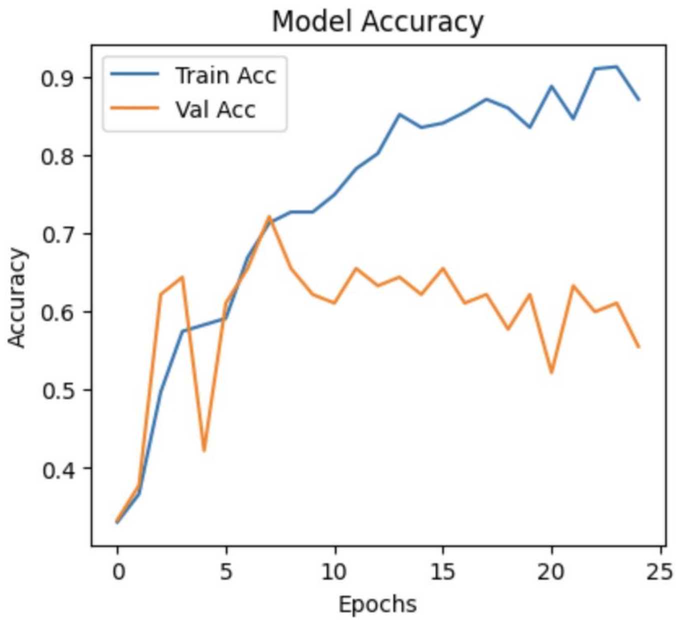
[12] OpenDrift v1.0: a generic framework for trajectory modelling. [16] A numerical oil spill model based on a hybrid method [17] Influence of sea surface waves on numerical modeling of an oil spill	Drift Prediction	Oil Spread Sim	Forecast future movements	Needs environment data

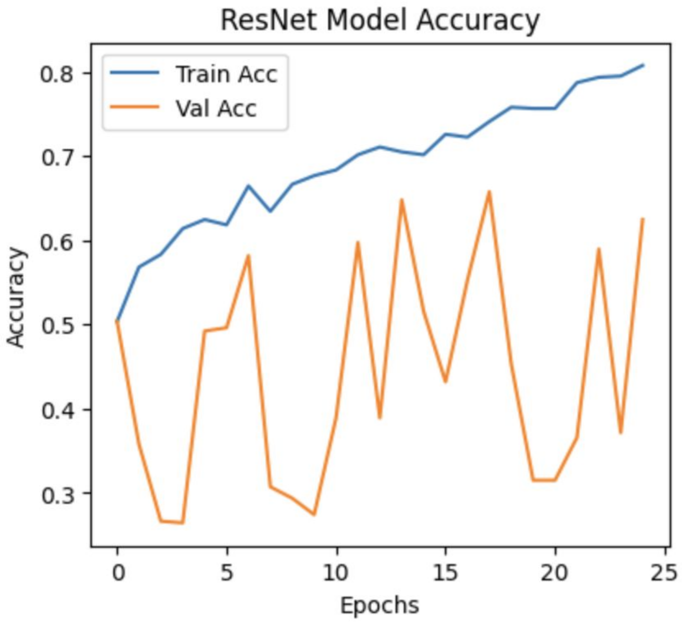
Methodology



Techniques Used

Detection Model Results





CNN

Validation Accuracy -> 55.56%
Training Accuracy -> 94.02%

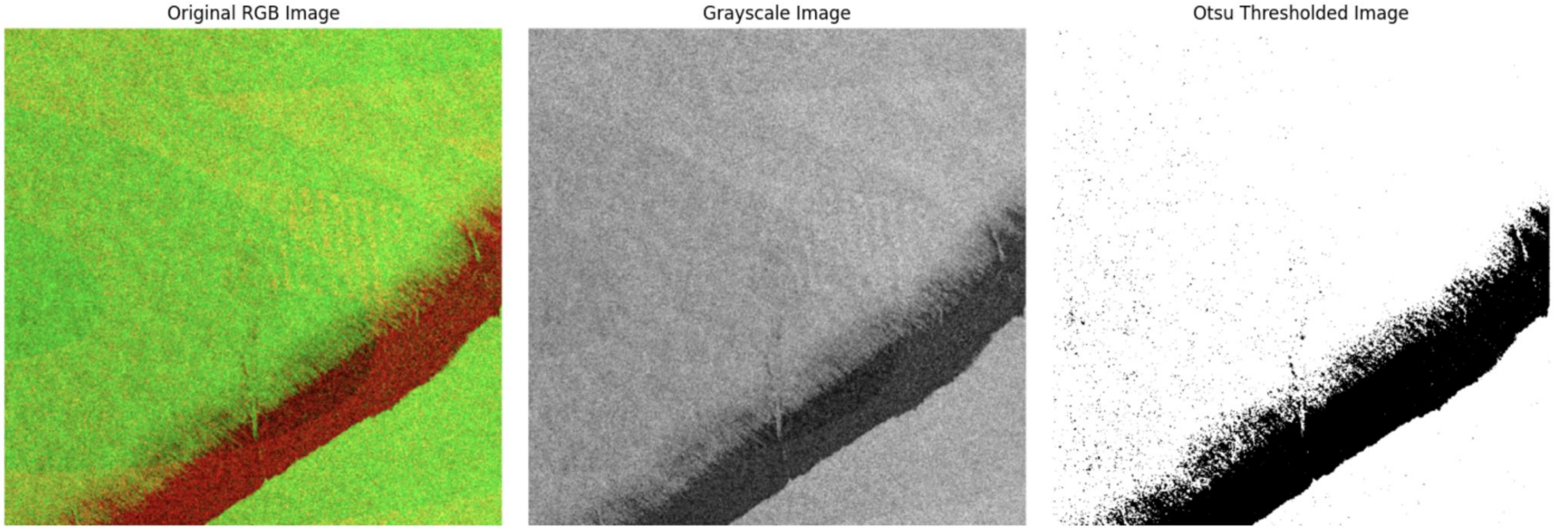
ResNet

Validation Accuracy -> 62.45%
Training Accuracy -> 80.74%

Image Segmentation

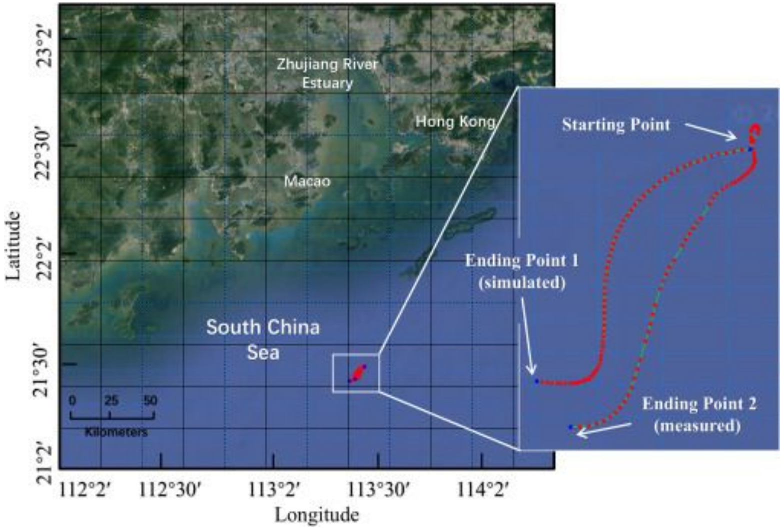
Otsu Thresholding:

- Otsu Thresholding (global thresholding) is an automated method for segmenting an image into foreground and background based on its intensity histogram.



Drift Prediction Techniques

Lagrangian Model



Wind drift parameter for lagrangian particle model of dummy drift trajectory in the Northern South China Sea

Lagrangian Models

- *Working:* Represent oil as many **virtual particles** that move with surface currents, wind drift, and waves.
- *Use:* Forecasts trajectory paths; intuitive for tracking oil parcels.

Eulerian Models

- *Working:* Solve **advection–diffusion equations** on a fixed **grid** to simulate oil concentration or thickness fields.
- *Use:* Captures spreading, weathering, and large-scale distribution over time.

Hybrid / Ensemble Models

- *Working:* Combine Lagrangian and Eulerian methods or run multiple simulations with perturbed inputs.
- *Use:* Provide probabilistic maps of possible slick locations, reducing forecast uncertainty.

Data Assimilation / ML Approaches

- *Working:* Integrate satellite detections (SAR, optical) or apply machine learning corrections to models.
- *Use:* Improves real-time tracking and prediction accuracy.

Modules Split Up

- **Module 1: Preprocessing Module** - Enhances and prepares SAR images through noise reduction, calibration, and segmentation for accurate analysis.

- **Module 2: Detection Module** - Detects and classifies oil slicks from SAR imagery using deep learning to distinguish spills, look-alikes, and clean water.

- **Module 3: Drift Prediction Module** - Predicts the movement of detected oil slicks using environmental and oceanographic parameters for timely response.

Module 1: Preprocessing Module

- Accepts raw **Synthetic Aperture Radar (SAR)** images from sentinel-1 API's, satellite datasets or repositories.
- Applies **noise reduction filters** to improve image clarity as images are pixelated.
- Normalizes and resizes all input images for consistent model training and inference.
- Extracts key **textural and statistical features** such as contrast.
- Outputs cleaned and enhanced SAR data for reliable oil slick detection.
- Models accuracy depends upon quality of preprocessing.

Module 2: Detection Module

- Accepts preprocessed SAR imagery from the previous module.
- Uses a **CNN** / other pretrained model to analyze spatial patterns in the imagery.
- Classifies regions into one of the three categories: **oil spill, look-alike,** or **no oil surface.**
- Generates **segmentation masks** highlighting the exact location and shape of detected slicks.
- Provides the essential input for drift prediction and visualization.

Module 3: Drift Prediction Module

- Accepts detection outputs and integrates **environmental parameters** such as wind speed, current direction, and temperature.
- Uses **drift modeling algorithms** (e.g., **Lagrangian / Eulerian / Hybrid models**) to simulate slick movement.
- Predicts the **trajectory of oil slicks over time.**
- Visualizes predicted drift paths on geospatial maps for better situational awareness.
- Serves as the final module providing actionable insights for **oil spill management and environmental protection.**

Proposed System

The proposed system is an entire pipeline designed to detect oil slicks and predict their drift using satellite and environmental data. The pipeline consists of :

1. **Input Module:**
   This module gathers data from the Sentinel-1 API (SAR imagery) and Weather API (wind, current, and temperature data). These inputs provide the essential raw data for analysis.

2. **Preprocessing Module:**
   It performs SAR preprocessing such as noise removal and calibration, and applies environmental data normalization to align satellite and weather information for accurate modeling.

3. **Detection Module:**
   This module conducts feature extraction to identify oil spill characteristics and executes detection algorithms to differentiate oil slicks from natural lookalikes in SAR imagery.

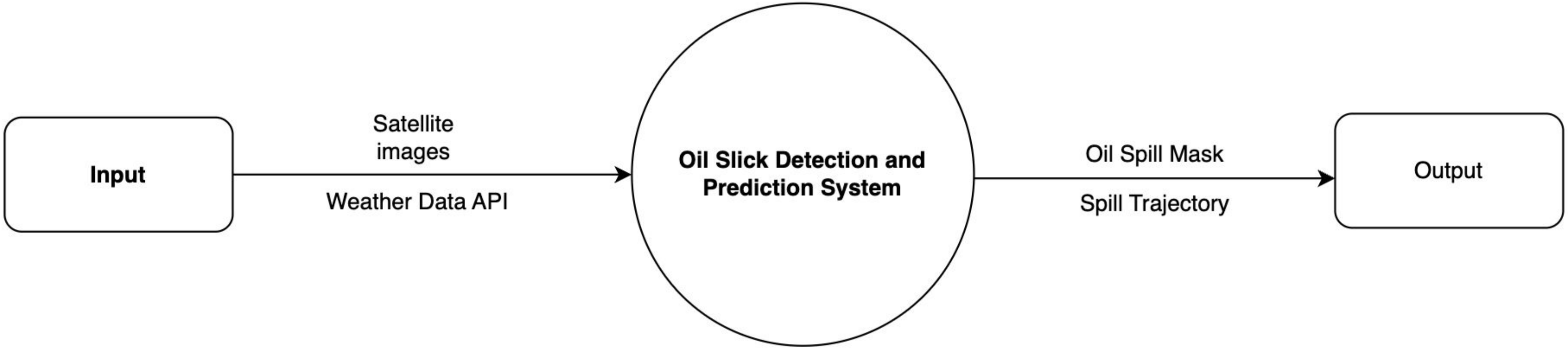
Proposed System

4. **Drift Prediction Module:**
   Using detected spill data and environmental inputs, the drift prediction model forecasts the movement and spread of oil slicks over time.

5. **Output:**
   The final outputs include an oil slick mask, showing detected spill regions, and a spill trajectory, visualizing the predicted drift path for monitoring and response planning.

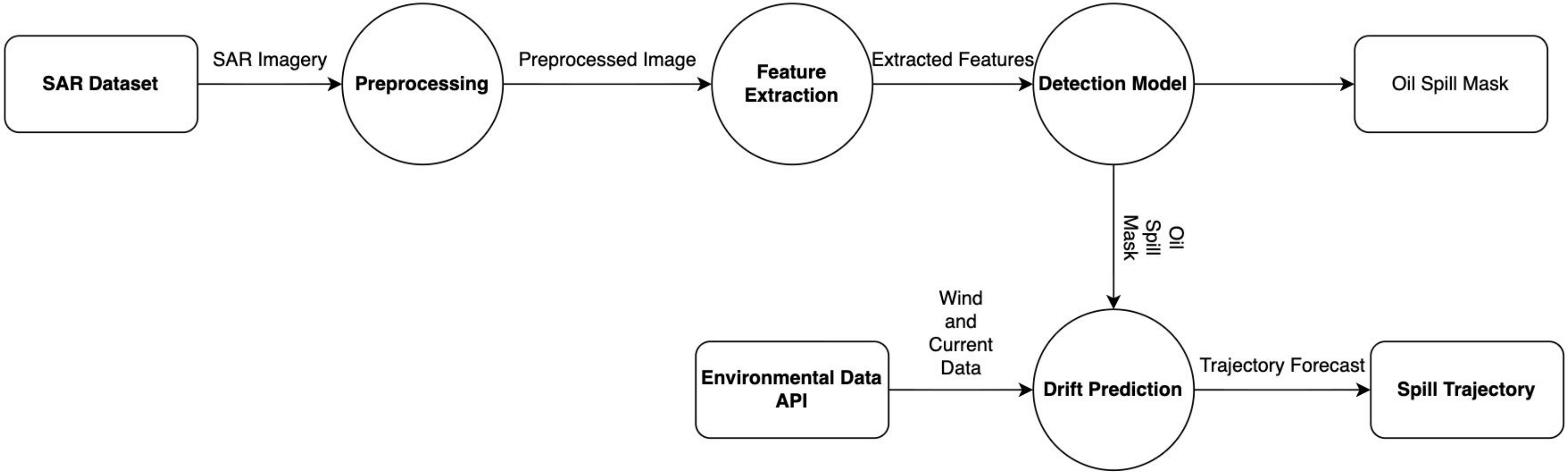
DIAGRAMS

Level 0 DFD



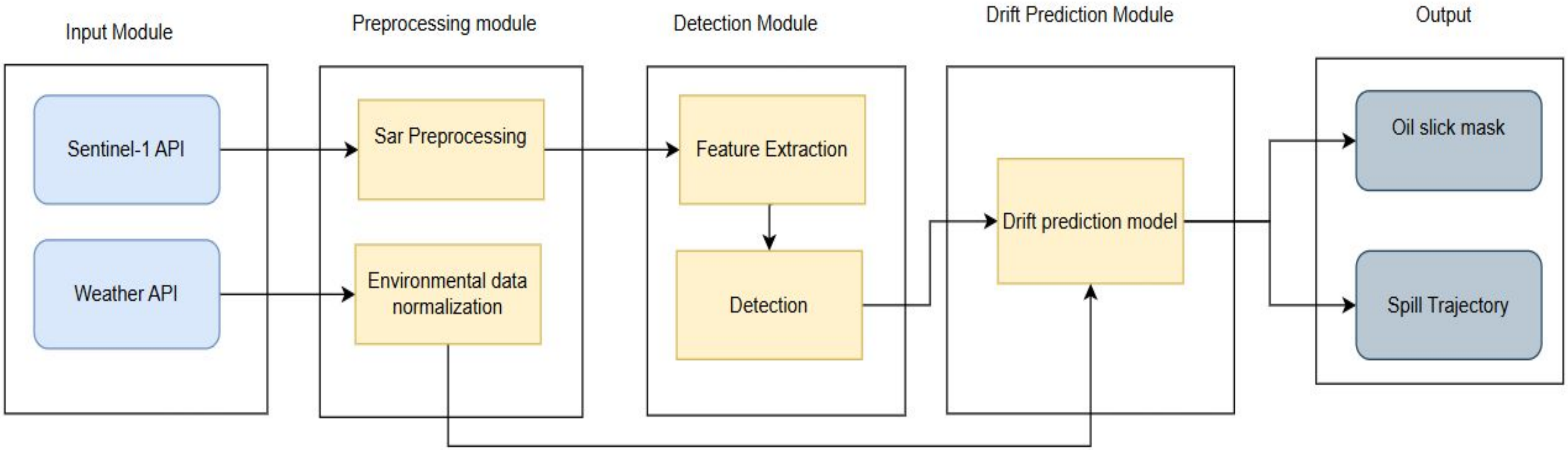
Input
Satellite images
Weather Data API
Oil Slick Detection and Prediction System
Oil Spill Mask
Spill Trajectory
Output

Level 1 DFD



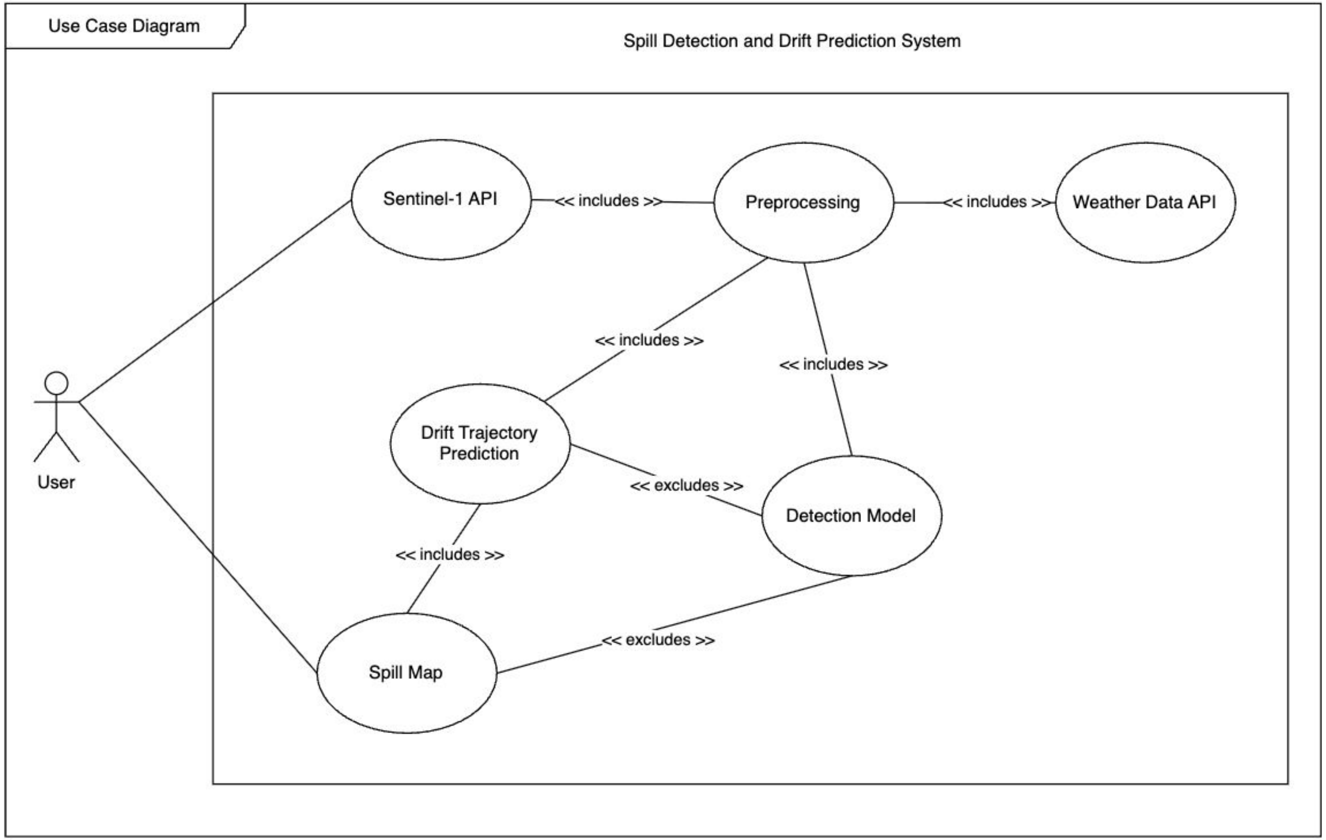
SAR Dataset
SAR Imagery
Preprocessing
Preprocessed Image
Feature Extraction
Extracted Features
Detection Model
Oil Spill Mask
Oil Spill Mask
Environmental Data API
Wind and Current Data
Drift Prediction
Trajectory Forecast
Spill Trajectory

Architecture Diagram

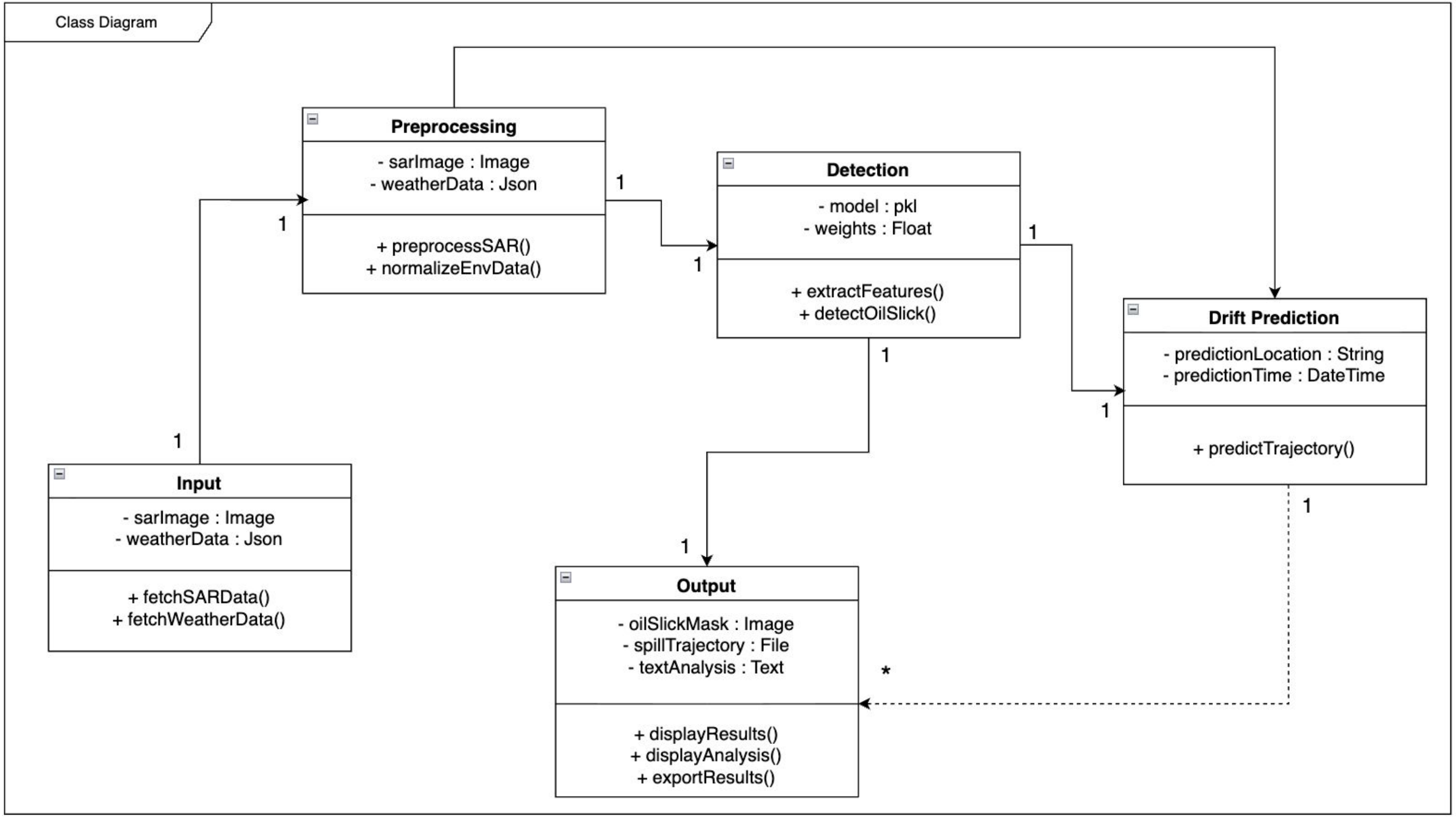


UML DIAGRAMS

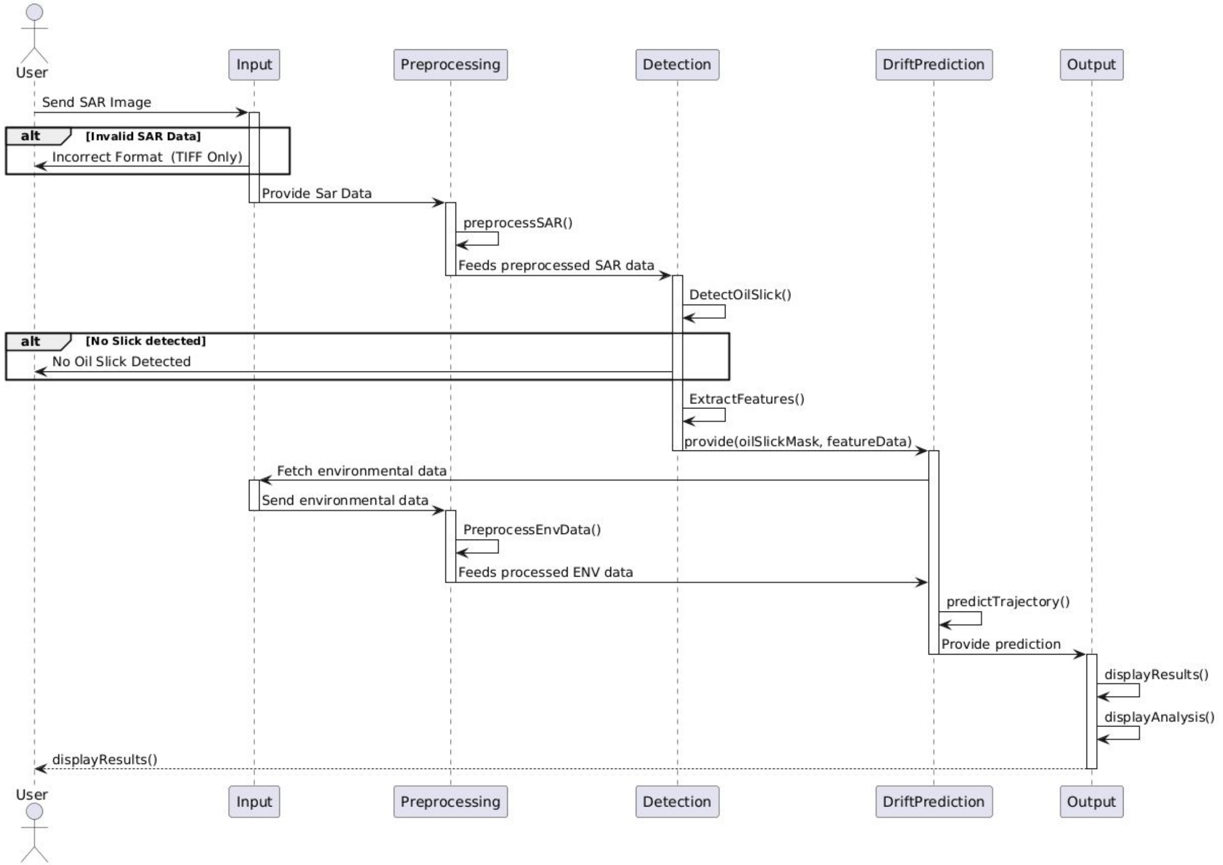
Use Case Diagram



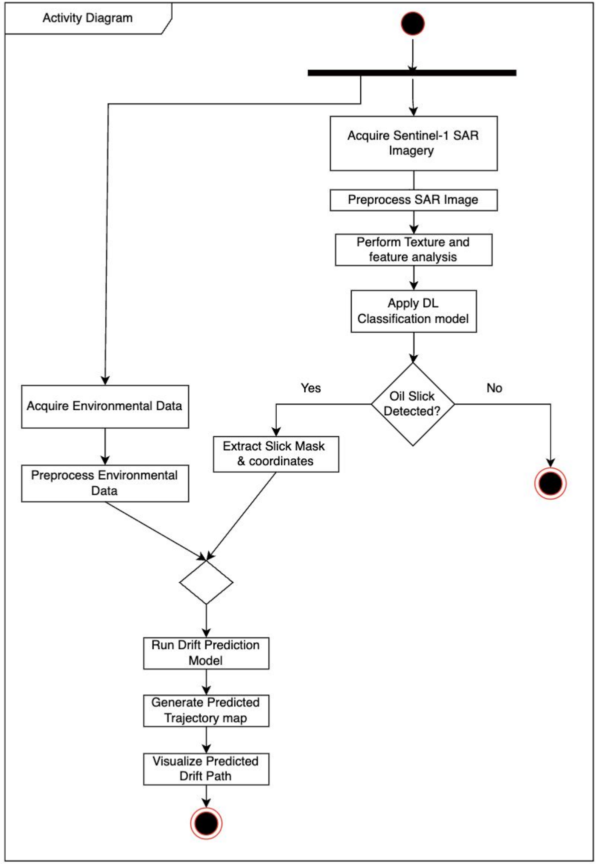
Class Diagram



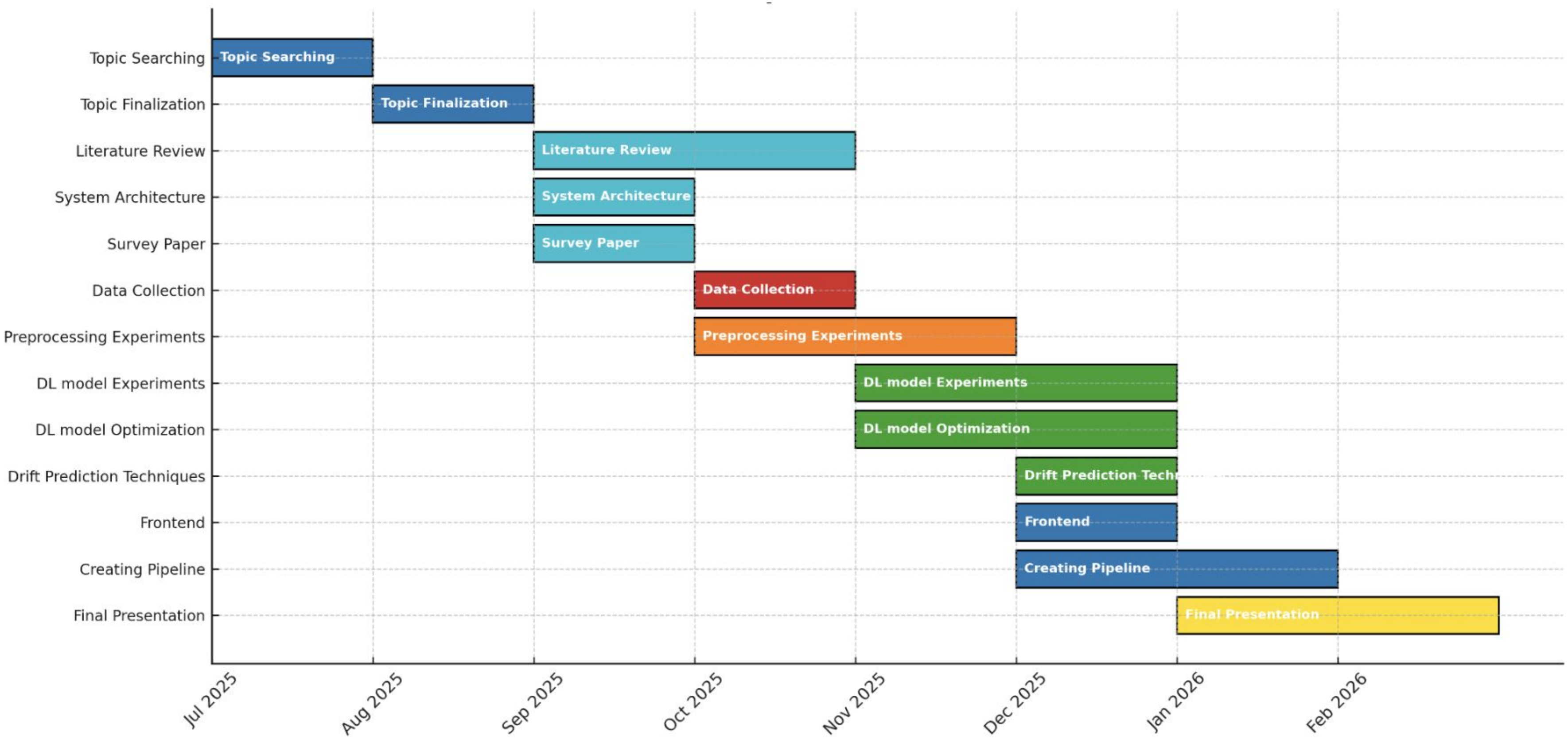
Sequence Diagram



Activity Diagram



Project Plan 2.0



Paper Acceptance

Hinweis Fourth International Conference on Artificail Intelligence and Data Science (AIDE)

Hybrid Mode; November 29-30, 2025

https://icaids.thehinweis.com/2025

PAPER REVIEW FORM	
Paper ID	**AIDE-2025_163**
Paper Title	OIL SLICK DETECTION and DRIFT PREDICTION USING SATELLITE IMAGERY
Review Status	**Accepted with Modification**
Category(if accept)	**Full Research Paper**
Consolidated Content review comments	• Paper should strictly formatted according to the standard format • Adequate testing, results and its discussion are required. The paper could be revised with more results • The paper could be revised with more detailed explanations to increase the understandability of the paper

Sl No	OVERALL RATING: (5= EXCELLENT, 1= POOR)	5	4	3	2	1
1.	Structure of the paper		X			
2.	Standard of the paper			X		
3.	Appropriateness of the title of the paper		X			
4.	Appropriateness of abstract as a description of the paper			X		
5.	Appropriateness of the research/study methods			X		
6.	Relevance and clarity of drawings, graph and table			X		
7.	Use and number of keywords / key phrases		X			
8.	Discussion and conclusion			X		
9.	Reference list, adequate and correctly cited				X	

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THANK YOU

Sequence Diagram